Concepção e Autoria Multimédia

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Project Development Phases

- Development
- Delivery
- Evaluation



Project Development Phases

- **Development**
  - The development phase of the multimedia project includes the **production and the testing**. Some development models treat production as a separate phase from testing.
  - The development phase starts after the design document has been evaluated, reviewed and signed-off by the client and other relevant stakeholders. This will usually have been facilitated by the use of a proof of concept to demonstrate that the design will deliver the 'right' product. The impact of changes to the design will have been noted and acknowledged by adjustments (signed-off by the client) to the project budget, project schedule and work breakdown structure.
  - In this development phase you will begin **prototyping, test and evaluate** each prototype as you build and integrate the product.
  - As a result of each stage testing and evaluation you will need to review the specifications and design before the next iteration of the development phase occurs.

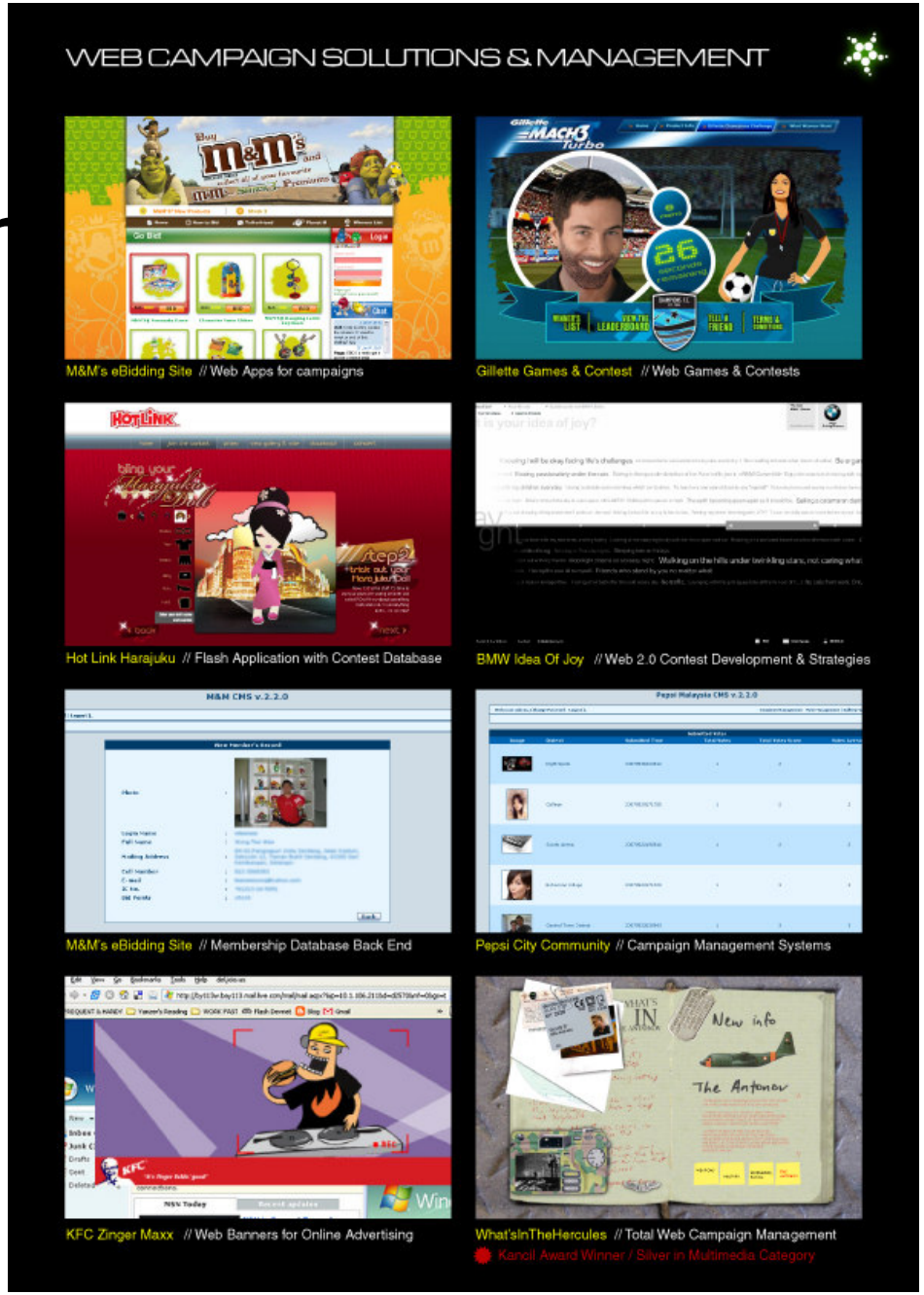
Project Developmer

- **Development - Prototyping**
  - A prototype of one or more aspects of a projected system is used to test the assumptions about the required system. The prototype may either form the basis of the next prototype and eventually the final product, or it may be discarded and the lessons learnt carried forward into the next iteration.
  - Adopting a prototyping approach is highly recommended, especially for large or complex packages. The **type of and number of prototypes will depend on factors such as size, complexity and skills and time available**.



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- **Development – Prototyping**
  - The reality is that most **multimedia products are developed through several iterations** of the phases of design (specification and design) and development (production and evaluation).
  - Different types of prototypes differ in what they model, how close they are to the real thing, who will 'use' them, what use they serve after the testing.



Project Development Phases

- **Development - Types of Prototyping**
  - A **horizontal** prototype is a shallow but wide model of the product, demonstrating many features and functions but mostly without detailed graphical treatment or functionality.
  - A **vertical** prototype models a narrow slice of the product with a few features developed with fairly detailed graphical treatment and functionality.
  - An **evolutionary** prototype forms a basis that is evolved into the next prototype, and after a number of such iterations, becomes the final product.
  - A **throwaway** prototype is discarded after evaluation and the lessons from it are carried forward into the next iteration.

Project Development Phases

- **Development - How Many Prototypes?**
  - For a relatively small and simple product, the proof of concept may serve as the single prototype with the final development then proceeding.
  - A typical prototyping approach and naming convention in software and multimedia production is to have an alpha prototype, followed by a beta prototype and then the final product.
    - The **alpha** prototype may be structurally fairly complete but has only a small part of the media content, as this is often still being developed. It can be used to clarify particularly difficult features or functions and/or the appropriateness of the programming tools. It is usually for the **purpose of testing within the team** and may not be provided to the client. However, if an electronic proof of concept was not developed an alpha prototype may provide the client with a first experience of the graphical treatment and functionality of the product.
    - The **beta** prototype usually contains the full structure, functions and content but may have bugs (documented and unknown) still to be resolved. Within the alpha and beta development stages various throwaways and evolutionary prototyping activities usually occur.

Project Development Phases

- **Development – Prototyping Strengths**
  - Prototyping recognises the **reality that the requirements of most software projects are very rarely fully known from the start** and **change over the life** of the development process. This approach allows for clarification of partly known requirements by quickly providing a working simulation, so that these can be tested and refined.
  - From the development and testing of the prototype, the developers **learn by doing**, and improve development strategies and techniques. The constant involvement of you and your users in testing the prototypes and clarifying the requirements leads to a developing sense of ownership of the system thus facilitating its eventual acceptance.

Project Development Phases

- **Development – Prototyping Weaknesses**
  - The role of the prototype may be misunderstood. Try not to let yourself or your users become too 'attached' to a 'good' prototype. Try not to object to changes or feel that a less 'successful' prototype was a waste of resources. Prototyping produces multiple versions of documentation, processes and products.
  - This environment **may be prone to a lack of control and poor adherence to reasonable standards**. Adapting to constant changes in requirements, and available technologies, may result in the project becoming bogged down in perpetual prototyping. Developers may also indulge in excessive iterative refinements. A threshold for accommodating such changes must be established.

Project Development Phases

- **Development - Build and Integrate**
  - The multimedia product is 'built' by producing the different digital media components (text, audio, video, and graphics) and combining these into an integrated multimedia product.
  - Support materials will be developed in parallel. It will be essential to ensure that references to the support materials from the multimedia package are accurate and up-to-date, and vice versa.
  - The different media components are developed in stages, with the design being refined and confirmed as the product is developed in progressive cycles.
  - Effective project management will coordinate these intensive production processes including overseeing an effective change control procedure including a file naming convention.

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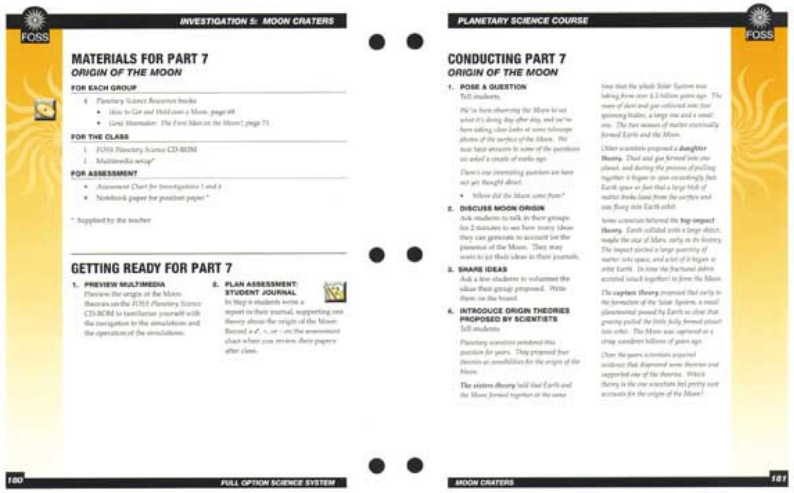


- **Development - Produce Media Components**
  - You will already have specified the types, quantity and quality of the media components to be produced during the design phase. Now in the prototype development of the product, the different media components will be developed in stages, with the design of each component being refined and confirmed as the product goes through progressive cycles.
  - The production of media components (text, graphics, video and audio) often involves specialists skills that may be provided by members of the core team or outsourced to individual contract workers or production organisations. You will need to verify that the media elements are produced according to the agreed specifications of quality and format.
  - Effective project management in this phase will coordinate these intensive production processes. A significant challenge is to adopt and follow an **effective file naming convention and change control procedure** to ensure that the correct version of a component can be located when needed. File naming and change control form part of configuration management, which will be determined and carried out as part of ongoing project management activities.

Project Development Phases

- **Development - Product**
  - Integrating the product involves combining the different media components with the graphical user interface and programming the interactivity between the different components.
  - Multimedia programmers/authors do not usually wait for all media components of the phase to be produced before developing the functionality and structure. Using **place markers and/or earlier versions of the media** they can develop the structure and program the interactivity whilst the media are still being produced. (The boundary between media production and integration is often blurred as many modern authoring tools actually have media production capabilities that allow the multimedia programmer/author to produce and integrate media within the same work environment and session)
  - To ensure effective progress towards the final product, changes have to be efficiently controlled by the change control and file naming techniques as set in the project's configuration management plan.

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- **Development - Support Materials**
  - You will already have decided what materials are going to be provided as support materials for the multimedia package.
  - These support materials should be developed in parallel with the multimedia product rather than after the development of the final multimedia product. This is necessary to ensure that references to the support materials from the multimedia package (and vice versa) are accurate.
  - The prototyping approach allows for content and design of the support material to be tested and evaluated along with the multimedia product. The parallel development of the support material also increases the likelihood of the project being completed on schedule and the quality of the support materials meeting your approval.

Project Development Phases

- **Development - Test**
  - The development phase ends with the **successful acceptance testing of the final product**. As the client you have to be diligent (and fair) to ensure that the product you are being asked to accept is indeed what you have requested through this iterative process of development and evaluation.
  - The **acceptance testing regime must be rigorous** enough for you to be fully satisfied as this is the product you will be getting for all your money and effort.
  - **Testing** involves checking a component or function to see that it meets a specified performance or quality. Testing is used to ensure that all built components and functions perform individually and as an ensemble according to the design intent. This will involve checking and 'debugging' the package to ensure that the code is strong and that the product works across all specified machines and browsers. The testing procedures form one part of the overall quality assurance of the product. Ample time should be set aside for testing the product and for correcting errors.

Project Development Phases

- **Development – Test**
  - **Determine the testing and evaluation techniques**
    It is best to determine the testing and evaluation techniques you are going to use for your project at the same time as you are developing the project plan. Otherwise, it is unlikely that you will have the time or the budget to undertake proper testing and evaluation. The evaluation and testing strategy can be documented in the project plan or in a separate evaluation and test plan. It may also form part of the project contract. Include information such as:
    - evaluation and testing methods
    - schedule
    - participants
    - reporting of results
    - procedure for sign-off and acceptance of the resource.

Project Development Phases

- **Development - Test**
  - Because of the iterative nature of the design and development process, many of the problems with the package will be uncovered and corrected as part of the process.
  - However, it is possible that items that function well as separate units will cease to do so in the fully integrated product. For this reason, a thorough testing of the integrated product of each cycle must take place.
  - Unfortunately, the chances of introducing a new error in the process of correcting an old one are fairly high. Decisions will have to be made about whether some errors are left 'unfixed' and documented as 'known problems' because of the implications that fixing the problem may have on the overall product.

Project Development Phases

- **Delivery**
  - The delivery phase of the multimedia project includes all the processes required to prepare and **deliver the final product and close the project**.
  - These activities follow on from the completion of the development phase, which has ended with the successful acceptance testing of the product. The processes involved in the close of the project include:
    - preparation for delivery (which includes archiving and debriefing the team)
    - delivery of the product.

Project Development Phases

- **Preparation for Delivery**
  - Preparing for delivery includes all the activities that '**freeze**' the final product and associated documentation and get them to the state where the product, if it is a CD-ROM, can be duplicated for delivery, or if it is a website, can be uploaded on the host server.
  - Once the products have been handed over, there is still the **need to sort and archive the project assets**. This is important for future product support, maintenance and upgrades.
  - There may also be components of the package that can be re-used for other projects. A project debriefing will allow you to review and document what went well and what went not so well on the project, so that your the organisation and the individual team members can benefit from the lessons to be learned.

Project Development Phases

- **Delivery – 'Freezing' the Product**
  - The activities will include:
    - freezing (no more changes allowed) and documenting the final list of what will be distributed as the final product files (including installation instructions and help files)
    - freezing and documenting the content and packaging of all support material that will be distributed with the product
    - burning the final program disk(s) (also called gold master disk) from which all copies will be replicated or uploading the website
    - checking that all files are present on the gold master disks.

Project Development Phases

- **Delivery - Archiving**
  - For the project team the project work is not finished at hand-over. There is often very valuable post-project archiving work to be done.
  - Once the products have been handed over, there is still the need to sort and **archive the multitude of project assets**, from multimedia files to project documentation. This is vital to ensure effective response to future product support and maintenance or future upgrades.
  - Many assets may have good reuse value (with due attention to copyright issues) or provide valuable information for future projects. Good directory management and file naming on the project facilitates archiving.
  - The time and resources for archiving have to be planned for early in the project schedule, or else there may be inadequate or chaotic archiving. Along with the project debrief, good archiving ensures that the organisation makes the most from the experience and outputs of the project.

Project Development Phases

- **Delivery - Project Debrief**
  - The purpose of project debriefing is to **review and document what went well and what went not so well** on the project, so that the organisation and the individuals may benefit from the lessons to be learned.
  - Although usually conducted with the core project team, the project debrief may also involve sub-contractors and even your representatives in certain sessions. A project review questionnaire may be circulated before the meeting to focus the team's thoughts on the subject.
  - Although problems should not be avoided, the challenge is to keep the exchanges positive with a **focus on "how can we do things better next time**?" The outcomes from the project debrief should be incorporated in a project report that covers the team and management's views on project performance (budget, schedule, team management, client satisfaction) and also documents the 'lessons learned'.

Project Development Phases

- **Delivery - Project Debrief**
  - Consider addressing the following questions in the project debrief:
    - To what extent did we achieve the **client's performance (quality) requirements**? (e.g., below target, on target, or above target).
    - How **close to budget was the final project cost**? (e.g., on target, overspent, underspent)
    - How did the **project completion date compare with the scheduled date**? (e.g., on target, behind, ahead).
    - What can we learn from the quality, cost, and time results in the above questions to apply in subsequent project initiatives?
    - What did we learn about the way we organized and managed the project?
    - What did we do very well on this project?
    - What could have been done better?
    - What have we learned that can be applied to future projects?

Project Development Phase



- **Delivery – Hand Over**
  - The delivery of the product entails the actual handing over of the final product of the project as stipulated in the project contract. If the product is a CD-ROM, this usually entails the gold master disk, the agreed number of CD copies and the support material. Sometimes it is negotiated with your developer whether they keep the gold master, or whether you receive the master copy. In many cases, the gold master remains the property of the development company.
  - The CD-ROM replication service to be used will have been identified and costed earlier in the project specification and design phase. Any particular standards required by the replication service would have been incorporated in the product specification.
  - If the product is a website, delivery involves the uploading of the site to the host server and handing over of support materials. The host server would have been identified earlier in the specification and design phase. Any particular standards required by the website hosting service would have been incorporated in the product specification.
  - This final product(s) should be accompanied by comprehensive documentation of all the files and items which make up the final products. The delivery is accompanied by the final sign-off, which indicates the end of the team's production responsibility to you.

Project Development Phases

- **Licensing, Copyright and Support Agreements**
  - Although the licensing agreement and copyright status of all materials will have been agreed to in the project contract, there may be need to reconfirm these as part of final signoff.
  - This may be particularly necessary if there were significant changes to the project scope and there are also materials or contribution of other parties involved.
  - Contractual obligations such as product warranty and ongoing support and maintenance should also be reconfirmed or may form part of a new contract. The copyright requirements of third parties should be confirmed and adherence to those verified.
  - **Note**: Using some development tools places certain requirements for mentions within the product credits and packaging. These should again be part of the delivery specifications developed in the specification and design phase, and ultimately verified as part of acceptance testing.

Project Development Phases

- **Evaluation**
  - Evaluation is an iterative process that focuses on the product's use rather than its features and functions. Done early enough it can discover design faults and reduce the costs of having to rebuild everything. It can reduce the need for a technical help desk and training costs, and can increase user satisfaction.
  - Evaluation can also reduce 'over design' by making you aware of what the user really needs. There is a range of different types of evaluation that can be applied to a multimedia product. This range includes but is not limited to the following:
    - usability
    - expert review
    - design evaluation
    - prototype evaluation.

Project Development Phases

- **Evaluation - Usability**
  - This evaluation assesses the effectiveness of the product design and its ease of use. It evaluates:
    - the product's 'learnability' (of the interface and functionality)
    - how easily the product allows users to recover from mistakes
    - whether the product is aesthetically pleasing and pleasant to use
    - whether users can accomplish from it what they want to.
    - It is important that any product that forms a vital component of a course is properly tested for its usability. In the design and testing of the product's usability, particular attention should be given to the needs of users with disabilities.The following sections outline some things to consider when evaluating your product.

Project Development Phases

Evaluation - Expert Review

- This involves the evaluation by multimedia experts and content experts (peers and external). They will provide feedback to the development team on all the aspects of product design, product content, products built (prototypes and final product), as well as project documentation.
- Expert review is one of the primary evaluation strategies used in both formative evaluation (i.e. how can this multimedia program be improved?) and summative evaluation (i.e. what is the effectiveness and worth of this multimedia program?).

Project Development Phases

- **Prototype Evaluation**
  - The evaluation of the multimedia product at each developmental cycle, verifies that the product has been built according to the design and validates that the product has been built according to your requirements. Testing will ensure that all components work individually and together as an ensemble. This is performed for the end product, the proof of concept and the alpha and beta prototypes. With the final product, this will be part of acceptance testing. Prototype evaluation is done by:
    - the project team
    - your representatives (with or without team members present)
    - a sample of end users on the alpha and beta versions, and
    - peer or external experts (with or without members of the project team).
  - Prototype evaluation is performed using evaluation instruments such as rating forms and questionnaires to evaluate aspects such as:
    - feature and structures
    - multi-platform and multi-environment performance
    - interface design
    - hardware and on-line performance
    - correctness of content.

Project Development Ph



- **Evaluating Utility**
  - Once you have your product, you will want to know how well it is received by the intended user, and how easy they find it to use your product. Evaluation techniques can include:
    - observation of people using the resource
    - interviews, focus groups and questionnaires
    - benchmarking and competitive analysis of existing products on the market
    - heuristic evaluations (evaluating software against accepted usability procedures)
    - hiring an external company who specialises in evaluation.
  - With results from these evaluations in hand, you may then choose to list ways of improving your product. You may decide to work on a second version or edition of the resource, based on your evaluations.

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